

Decision Making Under Uncertainty, MATH 20011

Homework 2: Hypothesis Testing

YOUR NAME HERE

2026-04-09

Instructions

- Replace “Your Name Here” in the author field above with your name.
- Follow the prompts in each part of this assignment and edit this .Rmd file directly. There are four parts to this assignment.
- Knit your finished document as PDF and submit the PDF on Canvas.

Libraries

```
library(openintro) # Library required to access our dataset
library(ggplot2) # Library required for visualization
library(dplyr) # Library required for data tidying and manipulation
```

PART 1: Data Exploration

We will use the `heart_transplant` dataset from the `openintro` package for this lab.

- (a) Load the specified data set. Hint: use `data()`

```
data(heart_transplant)
```

- (b) Compactly display the internal structure of the data. Hint: use `str()`

```
str(heart_transplant)
```

```
## tibble [103 x 8] (S3: tbl_df/tbl/data.frame)
##  $ id          : int [1:103] 15 43 61 75 6 42 54 38 85 2 ...
##  $ acceptyear  : int [1:103] 68 70 71 72 68 70 71 70 73 68 ...
##  $ age         : int [1:103] 53 43 52 52 54 36 47 41 47 51 ...
##  $ survived    : Factor w/ 2 levels "alive","dead": 2 2 2 2 2 2 2 2 2 2 ...
##  $ survtime    : int [1:103] 1 2 2 2 3 3 3 5 5 6 ...
##  $ prior       : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1 ...
##  $ transplant  : Factor w/ 2 levels "control","treatment": 1 1 1 1 1 1 1 2 1 1 ...
##  $ wait        : int [1:103] NA NA NA NA NA NA NA 5 NA NA ...
```

- (c) View the first 6 rows of your data. Hint: use `head()`

```
head(heart_transplant, 6)
```

```
## # A tibble: 6 x 8
##   id acceptyear age survived survtime prior transplant wait
```

##	<int>	<int>	<int>	<fct>	<int>	<fct>	<fct>	<int>
## 1	15	68	53	dead	1	no	control	NA
## 2	43	70	43	dead	2	no	control	NA
## 3	61	71	52	dead	2	no	control	NA
## 4	75	72	52	dead	2	no	control	NA
## 5	6	68	54	dead	3	no	control	NA
## 6	42	70	36	dead	3	no	control	NA

(d) Replace `eval=FALSE` with `eval=TRUE`. Explain the purpose and output of the following R code.

```
heart_transplant |> filter(!is.na(wait))
```

Question:

Based on your observations from the (a) - (d) and the information provided by (openintro)[<https://www.openintro.org/data/index.php>], describe the `heart_transplant` dataset. Your description must include, but not limited to:

- How many observations are in this dataset?
- What variables are included? Identify which are categorical and which are numerical.
- Why some entries in the `wait` column are NA?

Answer:

Part 2: Data Summary and Visualization

- (a) Generate a summary statistics of the data. Hint: use `summary()`
- (b) Create a bar chart showing the number of patients by treatment. Hint: use the `transplant` variable.
- (c) Create a histogram of patient ages. Hint: use the `age` variable.
- (d) Create a boxplot comparing survival time (`survtime`) by treatment (`transplant`) group.
- (e) Create a scatterplot of `age` vs. `survtime` with color as `transplant`.

Questions:

Based on your observation from (a) - (e), describe your data using the following prompts:

- The distribution of the age.
- The group (treatment/control) with the longest median survival time.
- Relationship between age and survival time.

Answer:

Part 3: Hypothesis Test for a Proportion

- (a) Create a binary variable indicating whether a patient survived more than 30 days post-transplant. Store the new tibble in a new tibble `heart_transplant2`
- (b) Replace `eval=FALSE` with `eval=TRUE`. Performing hypothesis test for one population proportion.

```
x <- sum(heart_transplant2$survtime_30)
n <- nrow(heart_transplant2)

prop.test(x = x, n = n, p = 0.5, alternative = "two.sided", conf.level = 0.95)
```

Questions:

- Explain each line of code within the code chunk in (b)

Answer:

- What is the sample proportion of patients who survived beyond 30 days?

Answer:

- Interpret the 95% confidence interval in context of the study.

Answer:

- State the null and alternative hypotheses for this test.

Answer:

- What is the p-value?

Answer:

- Do you reject or fail to reject the null hypothesis at $\alpha = 0.05$ and why?

Answer:

- Interpret the result of the hypothesis test.

Answer:

Part 4: Hypothesis Test for a Mean

In this part, we are interested in knowing whether the average age of heart transplant patients is different from 50 years?

Replace `eval=FALSE` with `eval=TRUE` to see the result from the code chunk.

```
t.test(heart_transplant2$age, mu = 50, alternative = "two.sided", conf.level = 0.95)
```

Questions: Provide answers to the following questions:

- Explain the line of code within the code chunk in this part

Answer:

- What is the sample mean age of the patients?

Answer:

- Interpret the 95% confidence interval in context of the study.

Answer:

- State the null and alternative hypotheses for this test.

Answer:

- What is the p-value?

Answer:

- Do you reject or fail to reject the null hypothesis at $\alpha = 0.05$ and why?

Answer:

- Interpret the result of the hypothesis test.

Answer: